

Read My Mind: EEG Motor Imagery Decoding with the Multi-Scale Dual-Axis Conformer

Walter J. Troiani[†], Josu Bernal[†]

Abstract—Brain-Computer Interfaces (BCIs) based on motor imagery (MI) offer a non-invasive way of translating neural intentions into external commands, offering significant potential for assistive technology. However, achieving high classification accuracy across diverse subjects remains challenging due to the inherent noise and variability of EEG signals. This study first reviews established methodologies in the field and subsequently proposes a novel framework: the Multi-Scale Dual-Axis Conformer. Unlike prior Conformer based architectures, our approach makes use of Multi-scale convolutional filters aligned with neurophysiological frequency bands ($\delta, \theta, \alpha, \beta, \gamma$) to enable this architecture to capture brain rhythms at multiple temporal resolutions and also explicitly models both temporal and spatial dependencies through dual tokenization processing of time-based and channel-based patch embeddings via self-attention and bidirectional cross-attention. Experimental results show our method outperforms current state-of-the-art models with t-SNE visualizations revealing more compact class clusters and clearer decision boundaries. However, superior accuracy comes at computational, with the architecture requiring approximately 4x more parameters than existing benchmarks due to the strong encoded inductive biases. A comprehensive ablation study revealed which components are redundant and which drive performance, providing insights into multi-scale convolutions and confirming the advantages of dual-axis tokenization, opening new research avenues.

Index Terms—Brain-Computer Interface (BCI), Motor Imagery, EEG classification, MultiScale-Dual-Axis-Conformer, Deep Learning.

I. INTRODUCTION

Brain-Computer Interfaces (BCIs) serve as a bridge between human neural activity and external devices, offering a communication channel. Among different modalities, EEG-based Motor Imagery (MI) is particularly relevant due to its non-invasiveness. Despite decades of research, the practical deployment of MI-BCIs often leads to poor generalization across different users or sessions. Traditional machine learning methods frequently struggle to capture the complex, non-linear dynamics inherent in noisy neural data. [1] [2]. While recent deep learning approaches have shown promise [3] [4] [5], the development of a unified framework remains an open challenge.

In this paper, we tackle the challenge of MI classification by proposing the MultiScale-Dual-Axis-Conformer, a model designed to capture brain signals across multiple frequency bands ($\alpha, \beta, \delta, \gamma, \theta$) while leveraging EEG-specific inductive biases through multi-scale convolutions. This conformer-based architecture (combining convolutions and transformers), inspired by [6] and [7], has been shown to outperform existing

state-of-the-art methods. By integrating specialized filters for different brain waves with a dual-stream attention mechanism, the model effectively learns both “when” and “where” critical neural activity occurs. Although highly effective, the model is computationally complex; however, the ablation study highlights areas for simplification and provides insights for future improvements along the Pareto frontier.

To summarize, the main contributions of this work are highlighted below:

- **Standardized Preprocessing and Augmentation:** We implement a comprehensive preprocessing pipeline, incorporating a low-frequency data augmentation strategy, that integrates multiple state-of-the-art techniques specifically tailored for EEG signals.
- **Systematic Literature Review:** We provide a detailed survey of MI-BCI methodologies, tracing the evolution of the field and conducting a comparative performance analysis on a common benchmark.
- **Novel Architecture and Validation:** We propose the MultiScale-Dual-Axis-Conformer, a novel architecture that establishes a new trajectory for EEG decoding research. Furthermore, we provide an extensive ablation study to identify and validate the specific components that drive the model’s performance.

The paper is structured as follows. In Sec. II, we review the related literature on motor imagery classification. Sec. III presents a high-level explanation of the pipeline we followed. Sec. IV, describes the dataset and preprocessing pipeline, while Sec. V details the proposed architectures and the conducted ablation study. The hardware details are attached in Sec. VI. The experimental results and performance evaluation are presented in Sec. VII. Finally, concluding remarks and future directions are provided in Sec. VIII. More information about this open-source research in [8].

II. RELATED WORK

Motor imagery EEG signal classification has evolved from traditional machine learning approaches, such as SVMs and ensemble methods, to fully automated deep learning frameworks [9]. Earlier pipelines generally involved three main steps: preprocessing, feature extraction, and classification. Preprocessing often included normalization, interpolation, ad hoc channel selection, or more general pipelines such as PREP processing [10]. Feature extraction focused on temporal, spectral, and especially spatial features [1], [2], which typically provided the best results. However, these approaches were sensitive to artifacts (e.g., heartbeats,

eye blinks), temporal dependencies, and required substantial biological expertise, making feature selection unstable and prone to human error. Recently, deep learning (DL) has emerged as the preferred approach, automatically learning complex features directly from raw data and overcoming the limitations of manual feature engineering.

Deep learning approaches to motor imagery classification follow distinct design philosophies. Some are inspired by traditional machine learning feature extraction. For instance, the Shallow ConvNet [5] and EEGNet [4] architectures are based on the Filter Bank Common Spatial Patterns (FBCSP) algorithm. Other methods adapt architectures from different domains; for example, [3] incorporates inception modules, originally developed for computer vision, into motor imagery classification. The multi-channel spatial structure of EEG signals makes convolutions particularly effective, while the temporal dynamics of EEG signals favor recurrent architectures. To capture both aspects simultaneously, hybrid approaches such as graph neural networks (GNNs), which utilize graph convolutions, and conformers provide naturally suited inductive biases for this problem. As noted in the Braindecode documentation [11], most state-of-the-art approaches can be categorized into one of these two families.

Evaluating deep learning methods fairly is challenging due to variations in datasets, problem scopes, and experimental setups. Additionally, each study is typically optimized for its own assumptions, making direct comparisons potentially misleading. In this work, we aim to provide a standardized comparison of the main approaches under the same conditions, while also reviewing some classical methods. Furthermore, we introduce our own Multi-Scale Dual-Axis Conformer, which builds upon previous findings [6] and leverages these insights to improve MI classification performance.

III. PROCESSING PIPELINE

Our pipeline begins with general-purpose preprocessing. Although deep learning architectures provide automated feature extraction and reduce the need for manual parameter tuning, we maintain a minimal preprocessing stage to standardize the input. We implemented a small pipeline inspired by PREP (standardized preprocessing for large-scale EEG analysis) [10], replacing robust referencing with Laplace re-referencing, as suggested by [12] and omitting the Line Noise Removal step since it has already been done by the data collectors. Laplace referencing enhances spatial locality and improves discrimination of sensorimotor rhythms in MI tasks. Next, we sliced the data into segments centered around the events, ensuring proper synchronization. Finally, we concluded the preprocessing phase with high-frequency data augmentation, inspired by [3].

Our processing pipeline diverges here. Although we mainly compared state-of-the-art deep learning models we decided to include some traditional approaches to give a broader

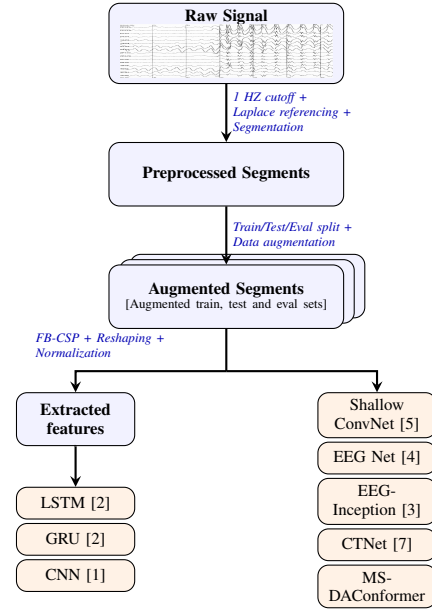


Fig. 1: High level overview of the complete preprocessing, feature extraction, and classification pipeline.

overview. For those models, features will have to be first manually generated, while the deep learning models will directly ingest raw, preprocessed signals. These features will be generated using FB-CSP to extract spatial-frequency sequential time slices, which will then be normalized. These features will be rearranged into various shapes suitable for different network types. The extracted features will then be classified by two different types of RNNs, a GRU, an LSTM [2], and a simple CNN [1], giving a total of 3 deep learning basic approaches and baselines to compare with. For advanced deep learning architectures, five approaches are studied: the Shallow ConvNet [5], the deep convolutional net presented in [4], EEGNet, an Inception module-based net as in [3], a CTNet [7], and our proposed architecture, the Multi-Scale Dual-Axis Conformer.

IV. SIGNALS AND FEATURES

We conducted our survey using a labeled EEG dataset divided into three distinct types of experiments, each with increasing levels of complexity. The first experiment, CLA, involves three classes: left hand, right hand, and a passive class. The second experiment, HaLT, expands this setup to include six classes: right leg, left leg, tongue, and the original hand classes. The third experiment, 5F, focuses on individual fingers, resulting in a five-class problem. This experiment is the most challenging due to the fine-grained nature of the movements involved. The dataset [13] is provided in MATLAB format and contains additional metadata that is not directly relevant to our survey, as all classification will focus only on the EEG recordings. Each recording consists of a two-dimensional array with dimensions $n_S \times 22$, where n_S represents the number of temporal samples, and 22 is the number of channels. The signal has mainly been sampled

at 200Hz, with some exceptions for the 5F experiment, where some data files have been recorded with a 1000 Hz sampling. The signals include 19 EEG channels arranged according to the 10-20 system, along with two reference electrodes (A1 and A2) and one synchronization channel (either X3 or X5). The signal measures voltage potentials as differences between an active electrode and a reference. The synchronization channel does not contain data; it is used to ensure synchronization between the EEG data and the corresponding event markers.

The preprocessing of the raw data involved three steps, with the first two being inspired by the PREP framework [10]. We began by applying a high-pass filter with a cutoff at 1 Hz, since complete line noise removal was unnecessary due to a 50 Hz notch filter already present in the data. Next, we used Laplace referencing instead of robust referencing, as suggested [12]. Re-referencing is to redefine the voltage reference against which all electrode signals are measured. Laplace referencing compares each electrode to a weighted average of its neighboring electrodes, instead of using a single global reference. After re-referencing the data, we applied a segmentation window around the marked events. The choice of window size was based on findings from [12], in which optimal window performance on the same dataset is studied. Next, we performed the usual Train/Test/Eval data splitting, and we also performed data augmentation on the training data to enhance robustness and reduce overfitting, improving both classification performance and stability. Our data augmentation approach [3] involves filtering out high-frequency components (above 90 Hz) from all samples and adding them into different samples, imitating how white noise data augmentation usually works, but using low frequencies as noise. Note that this step is performed only on the training data, to avoid data leaking. This process can potentially increase the dataset size by up to $n-1$; however, we found this unnecessary and only increased the data size by a factor of 8 for CLA and 5F, and by a factor of 3 for HaLT¹.

For spatial feature extraction, we employed the Filter Bank Common Spatial Pattern (FB-CSP) algorithm [14]. The FB-CSP framework consists of three primary stages. First, a filter bank is applied to decompose the EEG measurements into nine frequency bands. These bands are 4 Hz wide, starting from 4 Hz to 40 Hz (e.g., [4, 8] Hz, [8, 12] Hz, ..., [36, 40] Hz). This step can be computationally expensive. Second, the CSP algorithm is applied to each band to compute spatial filters. For each band, a transformation matrix W is calculated to project the band EEG data E , $Z = WE$. The goal of this matrix is to re-project the EEG signals into a new space where two classes are most distinct, by optimizing the variance of the resulting components. We compute this matrix using only the training set, to avoid leakage, but apply the transformation to

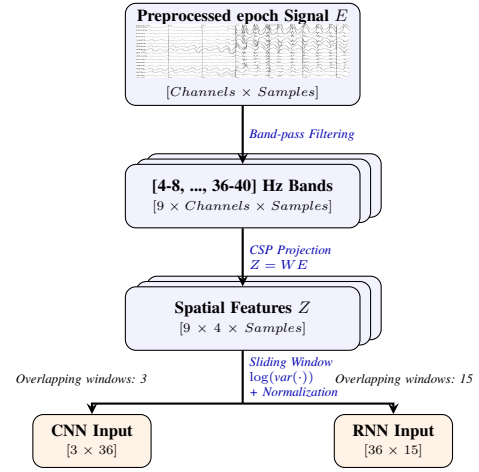


Fig. 2: Feature extraction diagram

all. To handle more than two classes, as we have, we employ a One-vs-Rest (OVR) strategy. Finally, we implemented a sliding window approach after spatial filtering. We extracted log-variance features from N overlapping temporal windows, and normalized them, creating a feature matrix suitable for CNN and RNN architectures, depending on how we arranged the computed features.

V. LEARNING FRAMEWORK

In this section, we go over the different models we used and explain briefly the main logic and intuition behind them.

A. Candidate Architectures

1) *Feature classification*: In this subsection, we will explain what models we used in the feature classification approach. We followed two different design viewpoints: a CNN, understanding time as “wideness of a picture” of features, and two RNNs. For both approaches, we windowed and reshaped our features accordingly. Firstly, we trained a common CNN, using 3 overlapping windows during the FB-CSP feature extraction algorithm, getting an input data tensor where each subject is a 3 by 36 matrix [1]. We compared this simple CNN approach with two different types of RNNs, to have an overall better understanding: an LSTM and a GRU. Both LSTMs (Long Short-Term Memory) and GRUs (Gated Recurrent Units) are types of RNNs designed to “remember” information, since standard RNNs tend to forget the beginning of a data sequence. They use internal “gates” to decide which information is important to keep and which can be discarded. LSTMs use three gates, while GRUs use only two. This simplification often results in faster training times and lower computational overhead. For this, we used 15 overlapping windows instead, resulting in a tensor where each subject is a 36 by 15 matrix. All this approach is inspired by [2].

2) *Signal classification*: The models in this section are more complex and require more explanation since they have to

¹The augment factor reduction was made due to technical RAM limitations

learn how to “extract the features” themselves automatically, feature learning.

a) *Shallow ConvNet (2017)*: It is a convolutional net, presented in [5], inspired by the FBCSP pipeline and designed primarily for oscillatory signal classification, for example, BCI-MI classification. The first two layers perform a temporal convolution and a spatial filter. After, a squaring nonlinearity, a mean pooling layer, and a logarithmic activation function follow; together these steps are analogous to the trial log-variance computation in FBCSP, properly learning the temporal structure of the band power changes.

b) *EEGNet (2018)*: EEGNet is a compact CNN architecture for EEG introduced by [4]. It has three main purposes: to adapt across different BCI paradigms, to work with very limited data, and to produce interpretable features. The model is divided in two blocks:

- **Block 1:** This block performs two sequential convolutional steps, inspired by FB-CSP algorithm mentioned before. The first is a convolution with the kernel at half the sampling rate, capturing frequency information from 2Hz and above. This is followed by a depthwise convolution to learn spatial filters. Other techniques complement the block: Batch Normalization, ELU activation function, Dropout for regularization, and Average Pooling to reduce dimensionality.
- **Block 2:** This block uses a Separable Convolution to efficiently process the extracted features. It consists of a Depthwise Convolution followed by a Pointwise Convolutions. This approach significantly reduces the total number of parameters and, decouples how a signal changes over time from how different signals relate to each other. First learns how to summarize individual signals in time and then chooses how to combine them. Finally, an Average Pooling layer is applied before the final classification. This approach allows EEGNet to generalize across different BCI paradigms, not necessarily Motor Imagery.

c) *EEG-Inception (2021)*: EEG-Inception, proposed in [3], ensembles Inception modules, adapted from computer vision, with residual connections. The Inception modules compute features from both the depth (series) and the width (parallel). To ensure stability, residual modules mitigate the vanishing gradient problem.

The Inception module contains a bottleneck layer, multiple convolutional layers, a pooling layer, batch normalization, and a RELU activation function. The bottleneck layer is a $[1 \times 1]$ kernel matrix that reduces the depth of the input data dimension, useful for feature reduction in computer vision. However, the layer can be inverted to increase the dimension and increase the number of trainable weights for deeper feature extraction. This inversion is applied only in the first layer; after that, the bottleneck layer is applied for decreasing the data dimension. The Inception module utilizes parallel convolutional layers to extract temporal

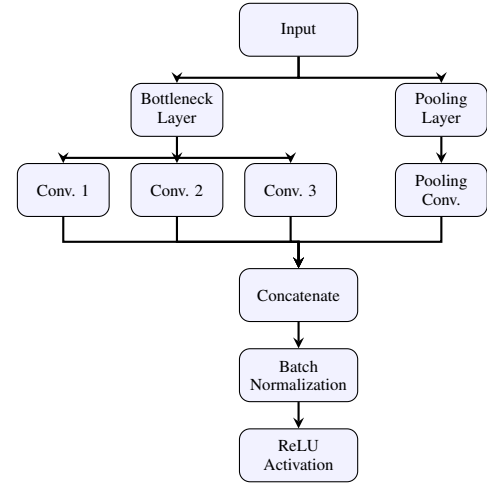


Fig. 3: Diagram of an Inception module with 3 parallel convolutional layers.

features across multiple time scales. By using different kernel sizes, the network can simultaneously capture information from both fast and slow oscillations in the signal. The pooling layer performs feature down-sampling by using a max pooling operation, selecting only the most prominent signal peaks. To compensate for this reduction in data length, a 1D convolution is then applied to increase the number of trainable parameters, allowing the model to learn more complex patterns from the signal.

d) *EEG-Conformer (2022)*: The EEG-Conformer proposed in [6] combines convolutional neural networks with Transformer encoders to leverage both temporal and channel-wise feature extraction. The convolutional front-end acts as a learnable feature extractor, replacing handcrafted feature pipelines, capturing short-term dependencies, and reducing dimensionality to form Patch Embeddings. Patch embeddings are fixed-length vector representations of short EEG segments, analogous to image patches in Vision Transformers, to avoid dealing with untractable complexity.

In this case, Patch Embeddings are produced using a ShallowConvNet-style convolutional stem in which temporal filtering is followed by global spatial mixing across all EEG channels. Subsequent temporal pooling aggregates the signal over short time windows, yielding fixed-dimensional tokens that summarize local temporal activity with globally mixed spatial information. While effective, this design implicitly entangles frequency and spatial representations and relies on a single temporal tokenization strategy. These patch embeddings or feature maps are then tokenized along the temporal dimension and passed to a Transformer encoder. Self-attention enables the model to capture global temporal dependencies across the entire EEG segment and avoid the RNN context bottleneck, see Fig. 4. Compared to pure CNN-based approaches, EEG-Conformer improves modeling capacity for long-range dependencies while maintaining a compact architecture.

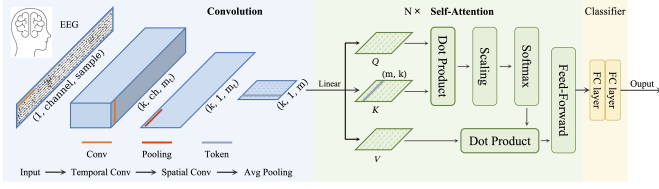


Fig. 4: Diagram of the EEGConformer architecture

3) Convolutional-Transformer-Network (CTNet) (2024):

The CTNet architecture proposed in [7] builds upon earlier convolution Transformer hybrids by refining the separation between local feature extraction and global temporal modeling. In contrast, CTNet adopts an EEGNet-style factorized convolutional embedding in which temporal filtering and spatial projections are explicitly decoupled. Depthwise spatial convolutions learn frequency-specific spatial filters, producing patch embeddings that preserve localized time-frequency-spatial structure prior to Transformer-based temporal modeling.

The architectural add-ons of CTNet are: positional encodings, which are added to presumably preserve temporal ordering of the patches, skip connections between the patch embeddings and the attention-enhanced embeddings, and more dropout to prevent overfitting. CTNet achieves a favorable trade-off between computational efficiency and representational capacity, and it's still considered a state-of-the-art architecture.

4) *Multi-Scale Dual-Axis Conformer* (2026): The proposed Multi-Scale Dual-Axis Conformer extends existing Conformer-based architectures by explicitly modeling EEG dynamics along both temporal and spatial axes and across multiple temporal resolutions. In contrast to standard EEG-Conformer models, which primarily apply attention along the temporal dimension, this architecture introduces two distinct token representations, time tokens and channel tokens, allowing attention mechanisms to operate explicitly along both axes.

The patch embedding front-end is a frequency-aware convolutional module with two key design principles:

- **Multi-scale temporal filtering.** Temporal features are extracted using convolutional kernels of different lengths, promoting an explicit multi-frequency inductive bias aligned with known EEG rhythms (e.g., δ , θ , α , β , γ bands). This design enables the model to capture neural dynamics at multiple temporal resolutions and produces richer patch representations than single-scale embeddings.
- **Dual-axis tokenization.** Instead of implicitly prioritizing the temporal dimension, the architecture constructs two complementary token types: (i) time tokens, obtained by aggregating information across channels to represent localized temporal segments, and (ii) channel tokens, obtained by aggregating temporal information per elec-

trode, preserving channel identity as a first-class modeling dimension that provides a different perspective. This design yields a strict superset of prior Conformer-based models, which rely exclusively on temporal tokenization and removes an inductive bias.

These two types of tokens are processed in parallel using positional encoding and multi-head self-attention ($(Q_t = T, K_t = T, V_t = T)$ and $(Q_c = C, K_c = C, V_c = C)$) applied independently to each token set. The model then enables joint representation through bidirectional cross-attention in both directions. In the channel-to-time pathway, channel tokens query time tokens ($Q = C, K = T, V = T$), answering the question *which temporal segments are most relevant for each channel?*. Conversely, in the time-to-channel pathway, time tokens query channel tokens ($Q = T, K = C, V = C$), addressing *which channels are most informative at each time step?*. This symmetric cross-attention design avoids introducing an inductive bias that prioritizes either the temporal or spatial dimension.

Finally, the attended representations are fused, flattened, and passed to a linear classification head. Dropout is applied throughout the architecture to mitigate overfitting. By jointly modeling multi-scale temporal structure and explicit spatial relationships, the Multi-Scale Dual-Axis Conformer provides a more neurologically grounded and expressive representation for complex EEG motor imagery.

B. Ablation study

In this subsection, we briefly explain how we performed an ablation study designed to isolate the contribution of each architectural component while keeping the remaining design fixed. The results will be discussed in Sec. VII. It aims to answer the following research questions:

- Are multiscale embeddings better than single scale? Is there a significant difference in the frequency range covered by these multiscale filters?
- Are both temporal and channel embeddings necessary, or does one dominate?
- Does cross-attention fusion provide additional gains over independent processing? Is it better not to use cross-attention at all?
- What is the optimal pooling strategy for aggregating dual-axis representations?
- How sensitive is the model to the number of attention layers and heads?
- Does positional encoding improve performance for EEG signals?
- How does the embedding dimensionality affect performance? Which is a clear bottleneck size?

With all these questions in mind we run the following experiments (A1, A2, B, C, D, E, F, G), leaving the rest of the architecture intact:

- 9 different experiments were ran, divided into two different groups that we called A1 and A2.

- **A1:** To determine if multi-scale convolutions provide gains over single-scale baselines with the same parameter number, we run the following experiments:

Metric	A1a	A1b	A1c	A1d
Kernel Sizes	(25,)	(64,)	(5, 25, 125)	(5, 12, 25, 64, 125)
Filters x Scale	(60,)	(60,)	(20, 20, 20)	(12, 12, 12, 12, 12)
Total Filters	60	60	60	60
Coverage	Single-scale	Single-scale	3-scale	5-scale

- **A2:** To test if kernel sizes aligned with neurophysiological frequency bands improve performance (Frequency Band Hypothesis, the whole premise under the inductive bias of these complex patch embeddings):

Exp ID	Kernel Sizes	Filters per Scale	Coverage
A2a	(3, 7, 15, 31, 63, 125)	(6, 8, 12, 12, 12, 10)	Full spectrum
A2b	(5, 12, 25, 64, 125)	(12, 12, 12, 12, 12)	Baseline
A2c	(7, 15, 31, 63)	(15, 15, 15, 15)	Mid-range only
A2d	(3, 7, 15)	(20, 20, 20)	High-frequency
A2e	(31, 63, 125)	(20, 20, 20)	Low-frequency only

- B) To test the contributions of temporal and channel embeddings:

Exp ID	Configuration	Purpose
B1	use 'time' embeddings=True, use 'channel' embeddings=False	Time tokens only
B2	use 'time' embeddings=False, use 'channel' embeddings=True	Channel tokens only
B3	use 'time' embeddings=True, use 'channel' embeddings=True, use 'cross' attention=True	Full model

- C) To test the contribution of all the permutations of cross attention:

Exp ID	Configuration	Purpose
C1	use 'cross' time' attention=False, use 'cross' channel' attention=False	No cross-attention
C2	use 'cross' time' attention=True, use 'cross' channel' attention=False	Time → Channel
C3	use 'cross' time' attention=False, use 'cross' channel' attention=True	Channel → Time
C4	use 'cross' time' attention=True, use 'cross' channel' attention=True	Bidirectional

- D) Next, different pooling strategies are checked:

Exp ID	Configuration	Purpose
D1	fusion' mode="flatten"	Preserve all spatial-temporal info
D2	fusion' mode="gap"	Global average + max pooling only
D3	fusion' mode="gap+flatten"	Hybrid approach

- E) 5 experiments were performed to understand the sensitivity of the model to the number of attention layers:

Exp ID	Num Layers	Time Self Attn.	Channel Self Attn.	Bi-Cross Attn.
E1	2	2	2	2
E2	4	4	4	4
E3	8	8	8	8
E4	16	16	16	16
E5	32	32	32	32

TABLE 1: Experimental configurations analyzing sensitivity to the number of attention layers.

- F) To check the effectiveness of the positional encodings:

Exp ID	Configuration	Purpose
F1	use 'positional' encoding=False	No positional encoding
F2	use 'positional' encoding=True	Positional encoding

- G) Finally, 5 more experiments were run to answer the impact of embedding dimensionality on performance and model size:

Exp ID	Embedding Size	Multi-Scale Filters	Bottleneck Ratio	Params
G1	20	(4, 8, 8, 8, 2)	0.67x	~0.6M
G2	40	(8, 16, 16, 16, 4)	0.67x	~1.0M
G3	80	(16, 32, 32, 32, 8)	0.67x	~1.8M
G4	128	(26, 50, 50, 50, 14)	0.67x	~2.5M
G5	256	(52, 100, 100, 100, 28)	0.67x	~4.2M
G6	384	(78, 150, 150, 150, 48)	0.67x	~6.1M
G7	512	(104, 200, 200, 200, 64)	0.67x	~8.0M
G8	768	(157, 302, 302, 302, 84)	0.67x	~11.5M
G9	1024	(209, 402, 402, 402, 112)	0.67x	~15.0M

TABLE 2: Experimental configurations analyzing the effect of embedding dimensionality on model capacity and parameter count.

VI. HARDWARE AND REPLICATION

Before we dive into the results, we consider it necessary to summarize the hardware utilized to run the experiments, since the computational power can be a differential factor when replicating these results. The characteristics are summarized on Tab. 3

Category	A100	L4
GPU Model	NVIDIA A100	NVIDIA L4
Architecture	Ampere (GA100)	Ada Lovelace
GPU Memory	40 GB (typical)	24 GB GDDR6
Host RAM (observed)	80 GB	53 GB
<i>Compute usage and cost assumptions</i>		
Credits per hour (Scenario A)	5.0 CU/h	1.7 CU/h
Cost per hour (Scenario A)	≈ 0.56 ¢/h	≈ 0.19 ¢/h
<i>Energy consumption and emissions (estimates)</i>		
Observed usage	3.0 h	24.45 h
Energy (low TDP case)	0.75 kWh (250 W)	1.76 kWh
Energy (high TDP case)	1.20 kWh (400 W)	1.76 kWh
Total energy (low case)		2.51 kWh
Total energy (high case)		2.96 kWh
CO ₂ (low case)		≈ 1.00 kg CO ₂
CO ₂ (high case)		≈ 1.18 kg CO ₂

Notes. Hardware specifications follow Google Colab public product documentation. Compute-unit consumption on Google Colab is not published as a fixed rate and may vary across accounts, regions, and time; Emissions assume an average electricity intensity of 0.4 kg CO₂/kWh and exclude data-center cooling overhead.

TABLE 3: Hardware characteristics, compute cost assumptions, and estimated energy footprint of GPU environments used in Google Colab

VII. RESULTS

In this section, we will only explain results that we find interesting. All experiment results can be found in the appendix; this way, we can focus on the important. In addition, bigger graphs are also presented there.

A. Architecture Comparison

In this subsection, we present the results of the different architectures for the three experiments we introduced before. All models have been trained under the same conditions and under the same data split. We use this preface to explain these shared characteristics: we train them all on 100 epochs, (but with a scheduler and low patience of 5 epochs) using the Adam optimizer, minimizing the categorical cross-entropy loss function, and F1 score as an evaluation metric. We performed all trainings using early stopping to avoid overfitting and class weights to deal with class imbalance. After the first comparison was done, we performed an ablation study on the model that we propose, showing how every part of our model contributes to the overall result. We decided to present only the results on the test set, and two metrics: Accuracy and weighted F1, and since we believe that an important factor to take into account is model complexity, we also compare the results with the total number of parameters of each.

B. CLA: 3 class experiment

CLA experiment is the experiment with fewer classes (left hand, right hand, and none), and therefore, we expect it to have the best overall results.

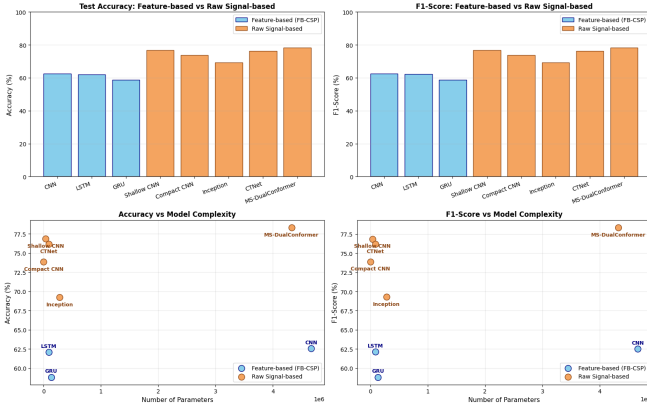


Fig. 5: Results of the CLA experiment

As expected, deep learning approaches consistently outperform traditional methodologies. Interestingly, between the feature-based architectures, achieving performance levels comparable to RNNs required a relatively complex CNN architecture. This is likely due to hyperparameter tuning during feature extraction optimization; however, given the high computational costs and our primary focus on advanced DL architectures, further optimization of these models was considered outside the scope of this survey. In addition, we want to underline that the Shallow ConvNet outperformed more complex models, probably due to its architecture being explicitly designed to compute power features. As demonstrated in later sections, this advantage disappears when the model is applied to more fine-grained tasks, which require capturing subtle amplitude variations and temporal potentials that the net's power-focused layers are

not optimized to do. Finally, while our proposed approach superiorly outperforms existing models, it does so with a significantly higher degree of architectural complexity.

We qualitatively compare the learned representations using t-SNE projections of the final-layer embeddings (Fig. 23). The Multi-Scale Dual-Axis Conformer exhibits more compact class-wise groupings and reduced overlap compared to CTNet, suggesting a more structured latent representation. Additional discussion and methodological details are provided in Appendix L.

C. HaLT: 6 class experiment

HaLT presents a scenario in which an increased number of classes increases the statistical complexity and dimensionality of the classification task, while the underlying physiological challenge remains consistent.

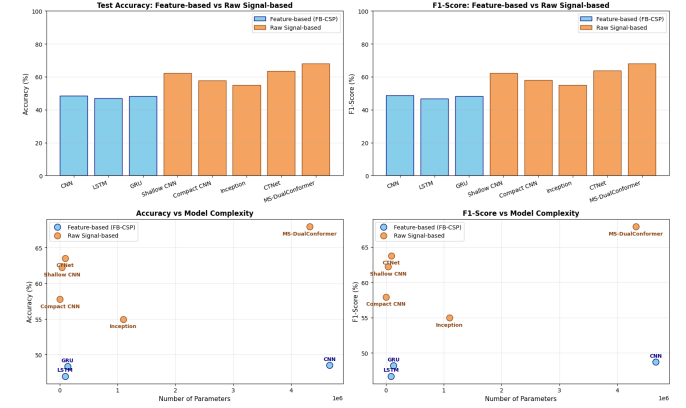


Fig. 6: Results of the HaLT experiment

The results are lower on average but present similar conclusions to the already presented ones. This lead us to assume that the intermodel performance relationship remains constant over the same physiological challenge, only reducing the accuracy due to higher dimensionality.

D. 5F: 5 class fine-grained movement experiment

5F presents a different challenge. Here, statistical complexity is lower than in HaLT, but the physiological similarities between finger movements are what make this challenge difficult.

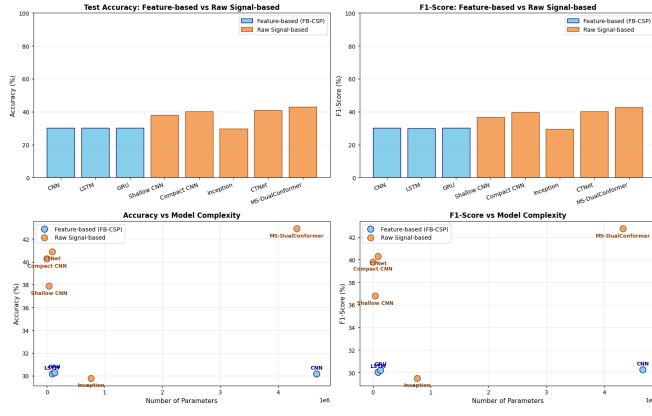


Fig. 7: Results of the 5F experiment

Even when performance exceeds randomness (20%), the overall results for this task remain poor. It should be highlighted how the Shallow ConvNet struggles to discriminate between classes in this problem, in contrast to previous tasks where it overperformed more complex architectures. This shows how physiological complexity yields the need for more robust and generalized architectures, like EEGNet.

E. Ablation results

Taking into account the favorable results of the MultiScale-Dual-Axis-Conformer, we believe that an ablation study could show interesting learnings that could lead to new research lines. Consequently, in this subsection, we show the results of the different experiments we ran during the ablation study. We will discuss the ones that we find revealing; all can be found on the appendix.

1) *A experiment: Are multiscale embeddings better than single scale? Is there a significant difference in the frequency range covered by these multiscale filters?:*

a) *A2 Experimental Group:* Surprisingly, the results showed that using single-scale embeddings has an overall better performance; there is no clear trend that using multiple-scale embeddings yields better results. This is probably caused by the bigger kernel sizes being able to capture multiple EEG rhythms at once. Hence, the hypothesis of the multi-scale patch embeddings should be revisited, because this could save a lot of model complexity in terms of parameters.

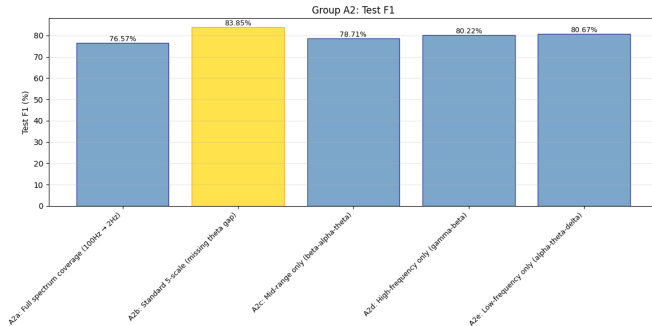


Fig. 8: Ablation A2 experiment

2) *B experiment: Are both temporal and channel embeddings necessary, or does one dominate?:* This result is especially interesting, since it basically describes the core motivation behind our architecture. As we hypothesize, using both temporal and spatial embeddings improves representation of the embeddings and the accuracy, and this is an optimistic result for research on this pathway.

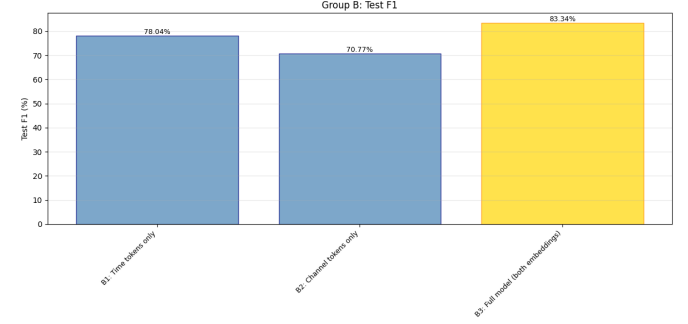


Fig. 9: Ablation B experiment

3) *C experiment: Does cross-attention fusion provide additional gains over independent processing? Is it better not to use cross-attention at all?:* Results are not clear, although it seems like time-channel cross-attention is the best, there is not a huge difference between any of the 4 options, so our initial hypothesis that both cross-attention directions are important does not seem to be true, but time-channel is the more relevant.

4) *D experiment: What is the optimal pooling strategy for aggregating dual-axis representations?:* Flattening the bidirectional cross-attention enhanced tokens seems to be the most performant method, although it has a clear trade-off with the number of parameters in the classifier MLP.

5) *E experiment: How sensitive is the model to the number of attention layers?:* It seems that a deeper attention network does not give better results immediately, but it could also be the case that with more training data this fact changes.

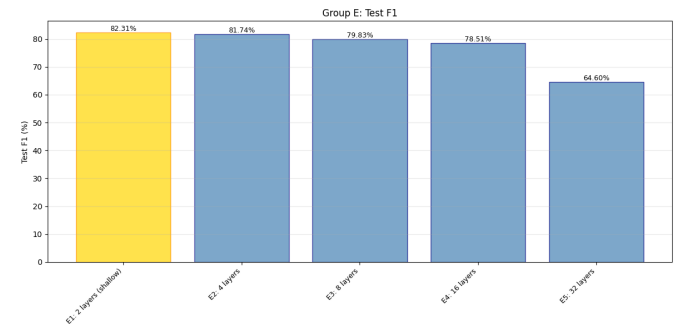


Fig. 10: Ablation E experiment

6) *F experiment: Does positional encoding improve performance for EEG signals?:* Surprisingly, results show again that positional encoding does not seem significant.

F. G experiment: How does the embedding dimensionality affect performance? Which is a clear bottleneck size?

The results for this experiment are unclear. Even if the largest embedding showed the best performance, the size increase does not seem to have a direct proportional influence on it. This led us to believe that size should not be the main concern after a certain threshold. Given that embedding size is the main bottleneck of this architecture complexity, we believe a small size, such as 20-64, falls in the ideal range. This is also sound bearing in mind that there are at most 21 electrodes in these datasets, making bigger dimensions probably degrade performance on channel embeddings, but potentially improving temporal embeddings, this is a great area for future work.

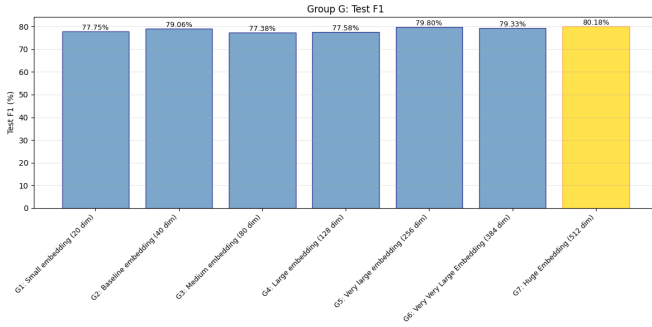


Fig. 11: Ablation G experiment

VIII. CONCLUDING REMARKS

A. Conclusions

This work benchmarked current state-of-the-art approaches to BCI-MI classification and proposed a new Conformer-based architecture, which outperformed the other methods. However, the model is considerably more complex than its competitors, although ablation studies showed that its size could be drastically reduced, representing a limitation that should be addressed in future work. The ablation study on the Multi-Scale Dual-Axis Conformer provided insights that can guide further investigation. In addition, it showed that some initial hypotheses were true (e.g., channel token identity provides a representational advantage), while others were found to be false (e.g., multi-scale patch embeddings did not provide a consistently better representation). Although we drew several insightful conclusions, the time-to-entry to the EEG community was high, which limited the extent of our experiments. Nonetheless, we believe this work presents publishable material with some polishing.

B. Future Directions

Despite its performance, the proposed model requires significantly more parameters than existing benchmarks, leading to computational bottlenecks. Future research should focus on:

- **Revisiting EEG-Conformer:** EEG-Conformer was excluded from the final comparison due to unstable training

behavior. Despite successful compilation of the TensorFlow computational graphs, we observed abnormal learning dynamics, including extreme or degenerate loss values. The source of this behavior is likely a subtle implementation issue within a specific layer or training component. Resolving this issue would allow EEG-Conformer to be included in future benchmarks, although preliminary evidence suggests it would likely underperform relative to CTNet.

- **Optimizing the Pareto Frontier:** Balancing model complexity and accuracy by simplifying the multi-scale components, optimizing embedding dimensionality (found to be a bottleneck in the 20, 64 range), and possibly restricting bidirectional cross-attention to a single time-to-channel pathway.
- **Enhanced Generalization:** Investigating why the 5F task remains challenging and exploring methods to better discriminate the subtle physiological similarities of individual finger movements.
- **Novel Architectures:** Exploring models beyond the Conformer paradigm, such as GNN-based Transformers that combine convolutions and attention, or Convolutional Spiking Neural Networks, which provide a novel and promising recurrent-native architecture despite its depth shortcomings.

C. Personal Remarks

Beyond the results, this project was a valuable exercise in academic research. We learned how to read and understand complex scientific papers and developed a critical eye when comparing different state-of-the-art methods. By analyzing these various approaches, we gained a better understanding of how research evolves. However, moving from papers to writing actual code revealed technical challenges. We ran into significant computational bottlenecks while extracting features, tuning hyperparameters, and training deep learning models. These tasks required so much processing power that we eventually had to pay for cloud computing services. Even so, time constraints and economic budget made us have to settle for a good but not optimal tuning of every model. On the other hand, this experience highlighted another common problem: scientific papers often prioritize theory over practical use or reproducibility. We believe that the research community would benefit greatly if papers were always accompanied by open-source code to bridge this gap.

REFERENCES

- [1] T.-J. Luo, C. Zhou, and F. Chao, "Exploring spatial-frequency-sequential relationships for motor imagery classification with recurrent neural network," *BMC Bioinformatics*, vol. 19, 09 2018.
- [2] B. E. Olivas-Padilla and M. I. Chacon-Murguia, "Classification of multiple motor imagery using deep convolutional neural networks and spatial filters," *Applied Soft Computing*, vol. 75, pp. 461–472, 2019.
- [3] C. Zhang, Y.-K. Kim, and A. Eskandarian, "EEG-Inception: An Accurate and Robust End-to-End Neural Network for EEG-based Motor Imagery Classification," *Journal of Neural Engineering*, vol. 18, 01 2021.
- [4] V. Lawhern, A. Solon, N. Waytowich, S. Gordon, C. Hung, and B. Lance, "EEGNet: a compact convolutional neural network for EEG-based brain-computer interfaces," *Journal of Neural Engineering*, vol. 15, 07 2018.
- [5] F. L. G. M. E. K. T. M. H. F. B. W. B. T. Schirrmester RT, Springenberg JT, "Deep learning with convolutional neural networks for EEG decoding and visualization," *Hum Brain Mapp.*, 08 2017.
- [6] Y. Song, Q. Zheng, B. Liu, and X. Gao, "Eeg conformer: Convolutional transformer for eeg decoding and visualization," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 31, pp. 710–719, 2023.
- [7] W. Zhao, X. Jiang, B. Zhang, S. Xiao, and S. Weng, "Ctnet: a convolutional transformer network for eeg-based motor imagery classification," *Scientific Reports*, vol. 14, 08 2024.
- [8] eZWALT, "Read my mind: Eeg motor imagery decoding with the multi-scale dual-axis conformer." <https://github.com/eZWALT/ML4HD-Machine-Learning-for-Human-Data/tree/main/PROJECT>, 2026. Commit / version used: main.
- [9] H. Altaheri, G. Muhammad, M. Alsulaiman, S. Amin, G. Altuwaijri, W. Abdul, M. Bencherif, and M. Faisal, "Deep Learning Techniques for Classification of Electroencephalogram (EEG) Motor Imagery (MI) Signals: A Review," *Neural Computing and Applications*, vol. 35, 08 2021.
- [10] N. Bigdely-Shamlo, T. Mullen, C. Kothe, K.-M. Su, and K. A. Robbins, "The PREP pipeline: standardized preprocessing for large-scale EEG analysis," *Frontiers in Neuroinformatics*, vol. 9, 2015.
- [11] B. Aristimunya, P. Guetschel, M. Wimpff, L. Gemein, C. Rommel, H. Banville, M. Sliwowski, D. Wilson, S. Brandt, T. Gnassounou, J. Paillard, B. Junqueira Lopes, S. Sedlar, T. Moreau, S. Chevallier, A. Gramfort, and R. T. Schirrmester, "Braindecode: toolbox for decoding raw electrophysiological brain data with deep learning models," 2025.
- [12] Y. Mishchenko, M. Kaya, E. Ozbay, and H. Yanar, "Developing a 3- to 6-state EEG-based brain-computer interface for a robotic manipulator control," *BioRxiv*, vol. 9, 08 2017.
- [13] M. Kaya, M. Binli, E. Ozbay, H. Yanar, and Y. Mishchenko, "A large electroencephalographic motor imagery dataset for electroencephalographic brain computer interfaces," *Scientific Data*, vol. 5, p. 180211, 10 2018.
- [14] H. Z. Kai Keng Ang, Zheng Yang Chin and C. Guan, "Filter Bank Common Spatial Pattern (FBCSP) in Brain-Computer Interface," in *2008 International Joint Conference on Neural Networks (IJCNN 2008)*, (Hong Kong, China), 2008.

APPENDIX

A. CLA

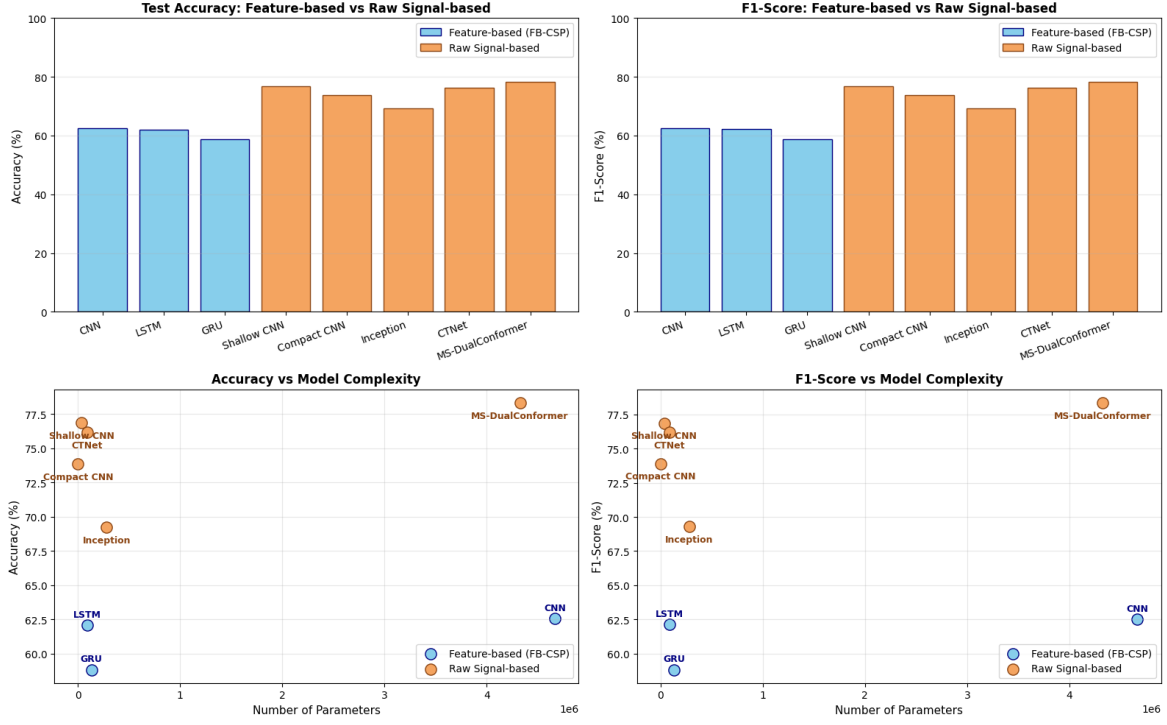


Fig. 12: Results of the CLA experiment

B. HaLT

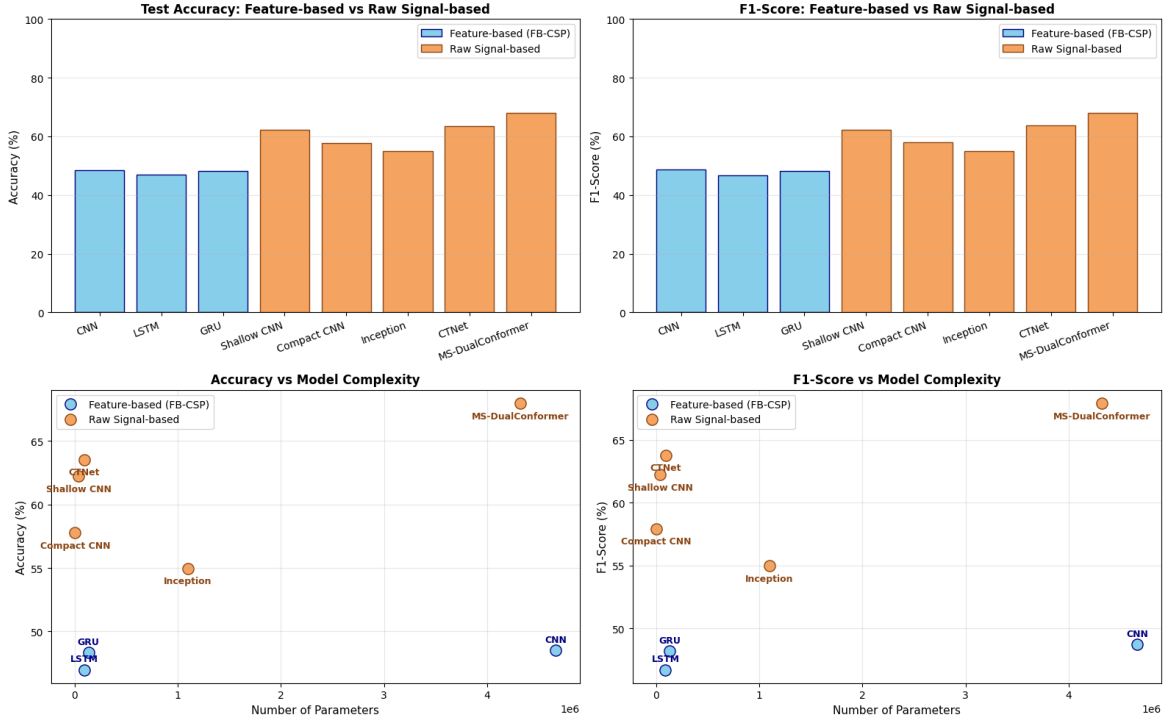


Fig. 13: Results of the HaLT experiment

C. 5F

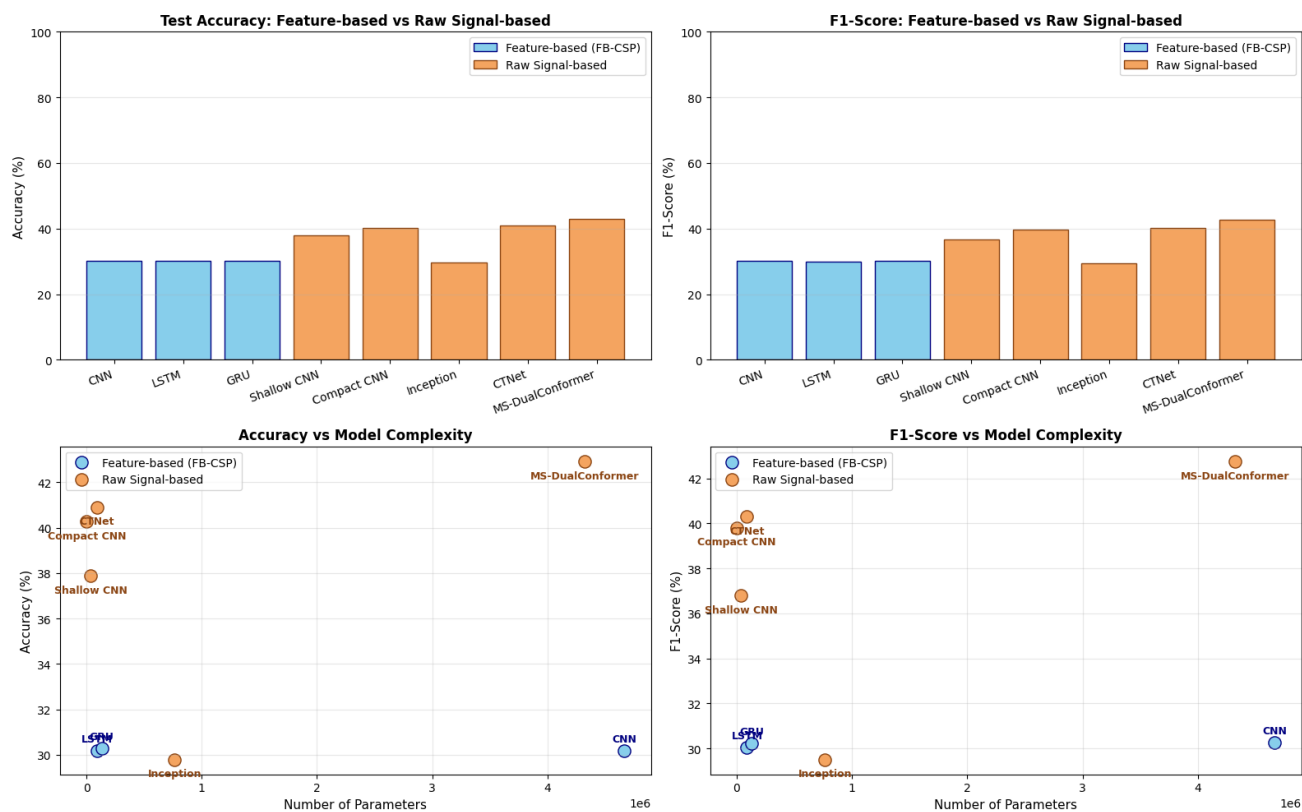


Fig. 14: Results of the 5F experiment

D. A1 Experimental Group

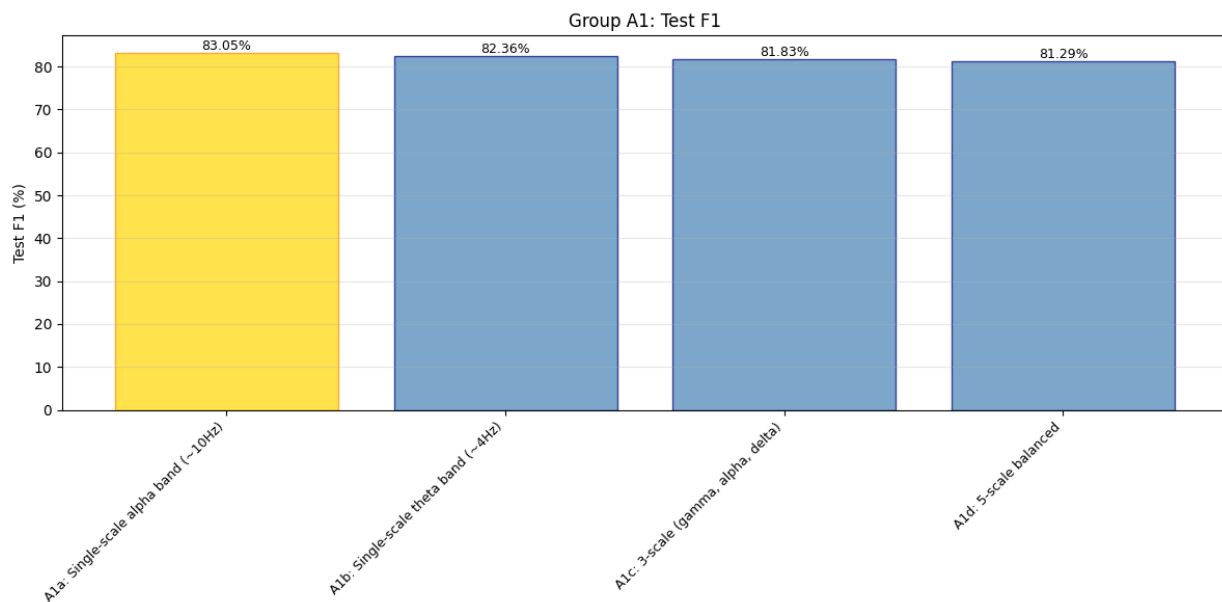


Fig. 15: Ablation A1 experiment

E. A2 Experimental Group

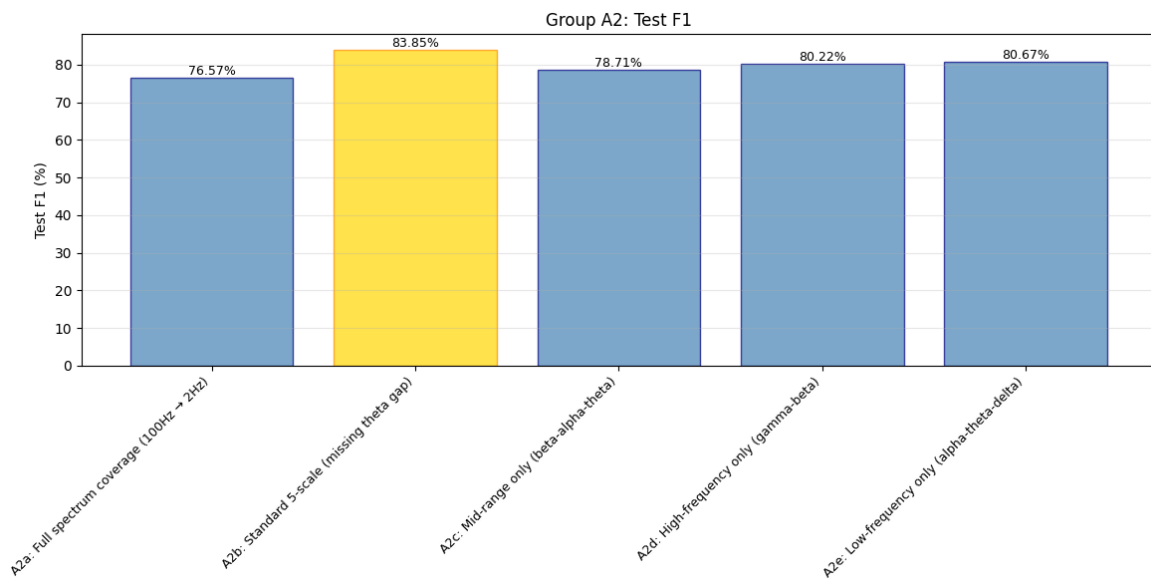


Fig. 16: Ablation A2 experiment

F. B experiment

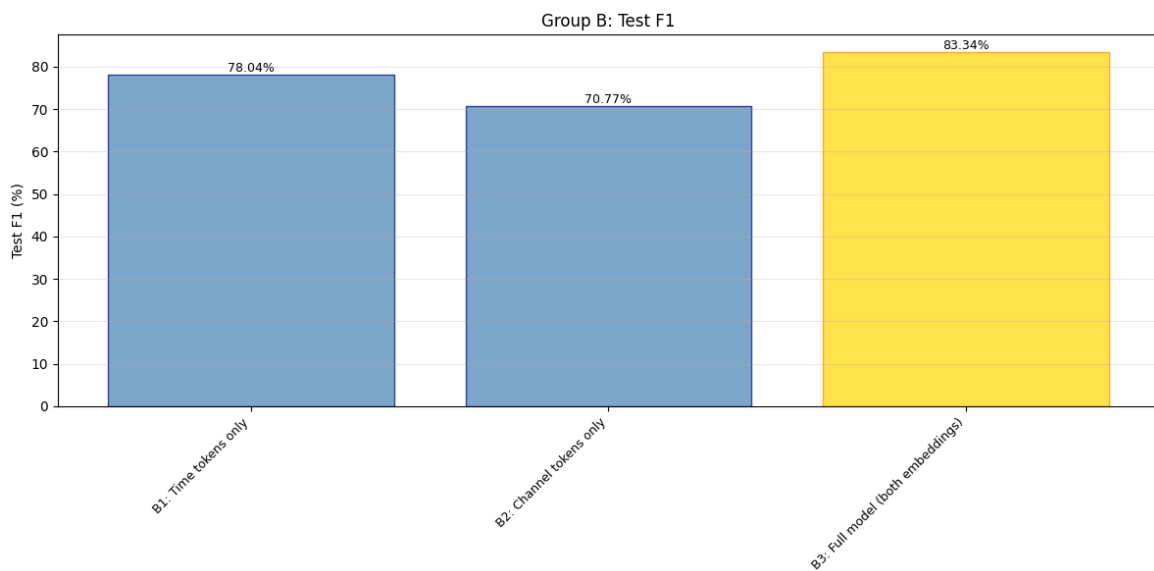


Fig. 17: Ablation B experiment

G. C experiment

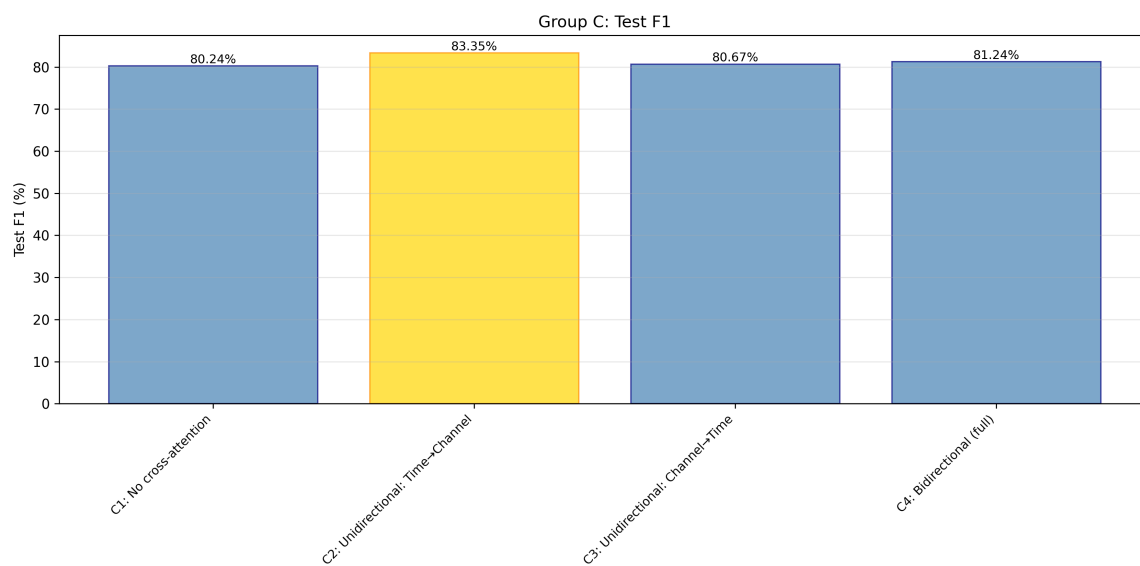


Fig. 18: Ablation C experiment

H. D experiment

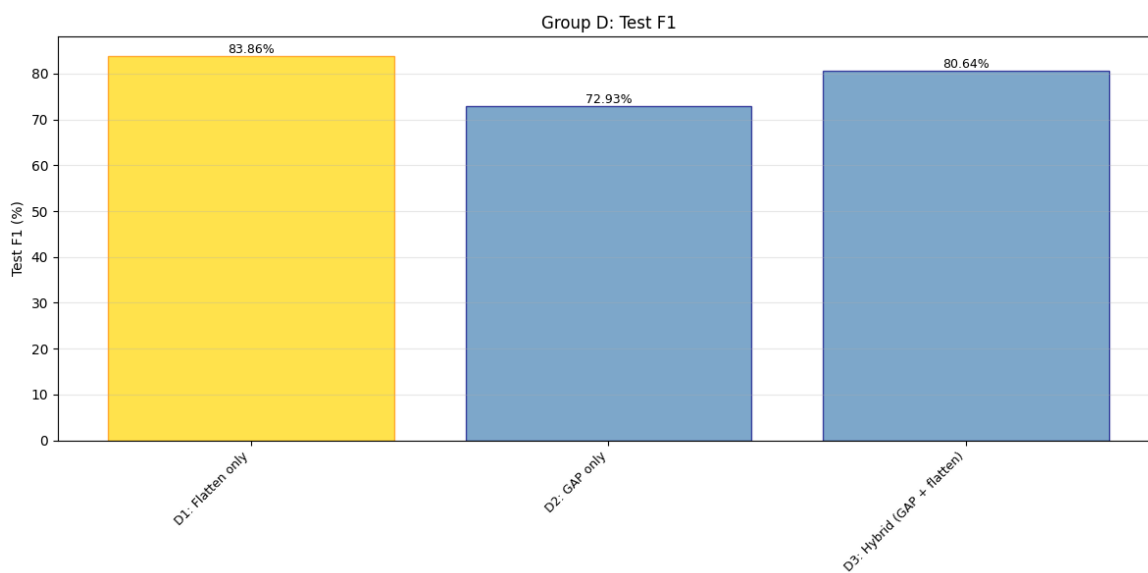


Fig. 19: Ablation D experiment

I. E experiment

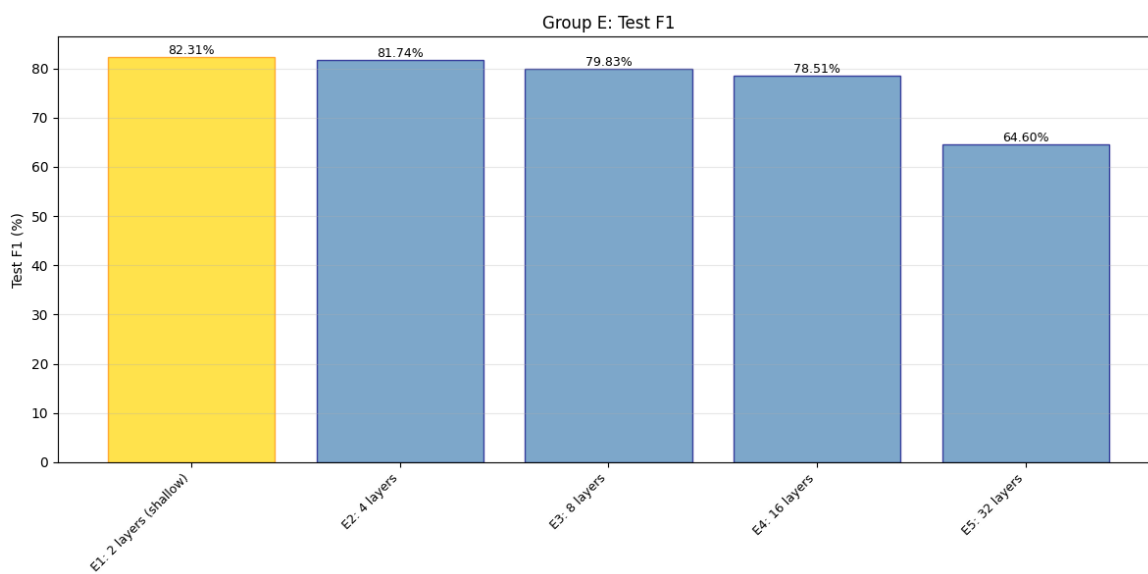


Fig. 20: Ablation E experiment

J. F experiment

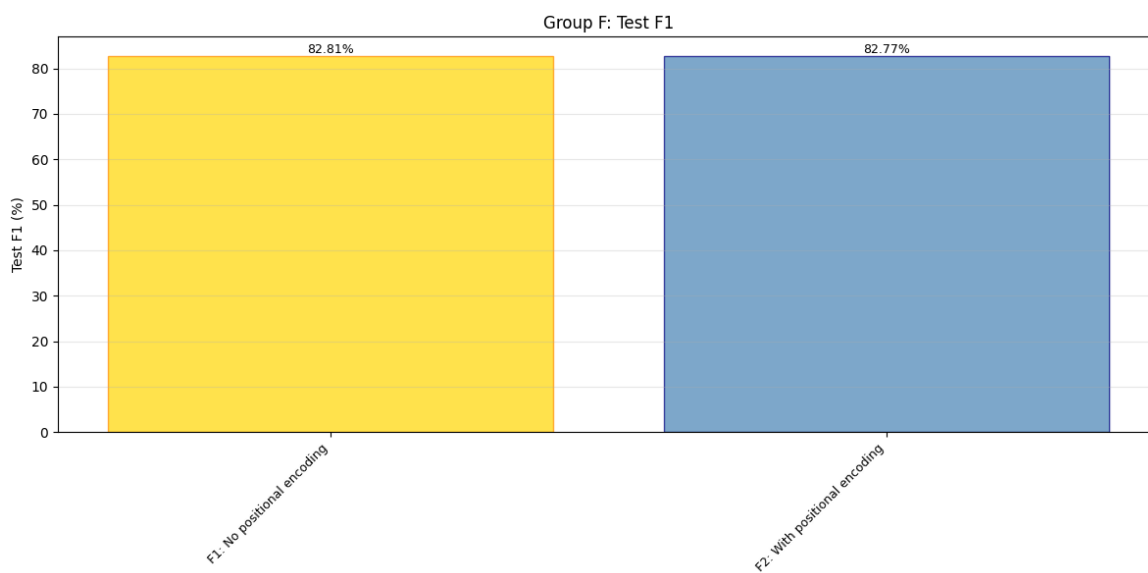


Fig. 21: Ablation F experiment

K. G experiment

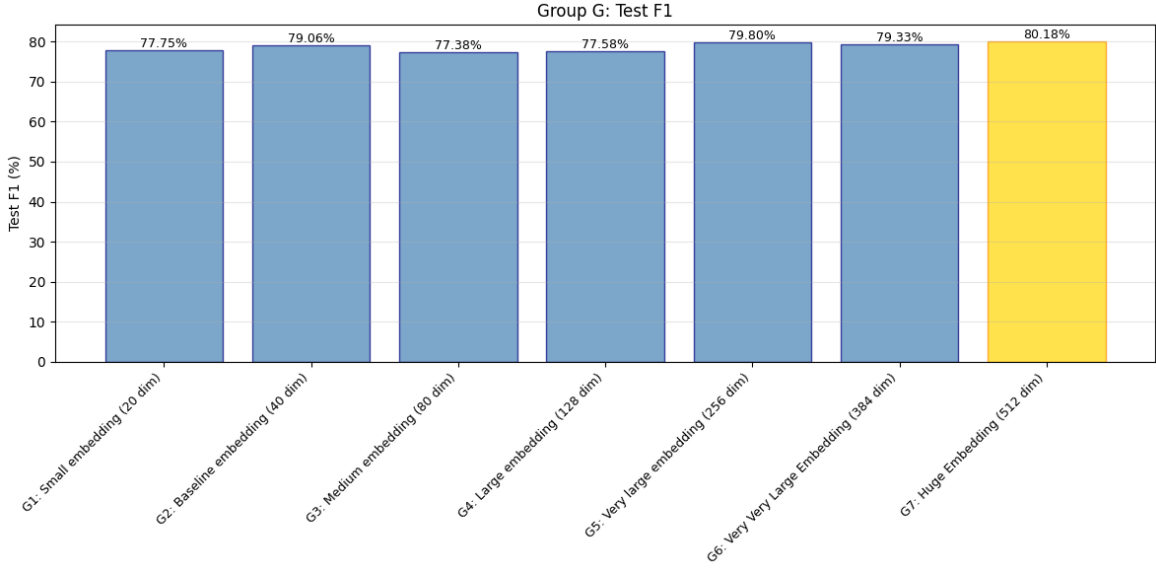


Fig. 22: Ablation G experiment

L. T-SNE Latent Space Comparison

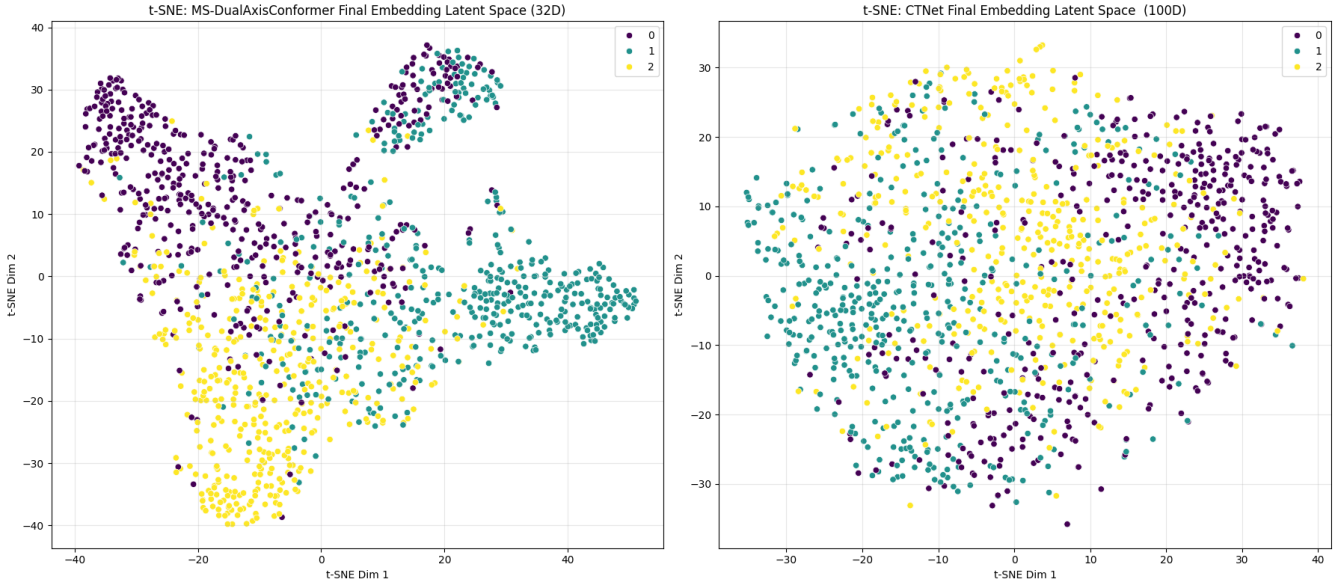


Fig. 23: t-SNE final embedding latent space comparison, MS-DA-Conformer (Left) vs CTNet (Right)

To gain further insight into the structure of the learned feature space, we apply t-Distributed Stochastic Neighbor Embedding (t-SNE) to the embeddings extracted from the final linear layer of each model. t-SNE is a nonlinear dimensionality reduction technique designed to preserve local neighborhood relationships rather than global geometry, making it suitable for qualitative comparison of latent representations rather than quantitative assessment of class separability.

Given high-dimensional embeddings $\{\mathbf{x}_i\}_{i=1}^N$, t-SNE defines pairwise similarities in the original space as conditional probabilities

$$p_{j|i} = \frac{\exp(-\|\mathbf{x}_i - \mathbf{x}_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|\mathbf{x}_i - \mathbf{x}_k\|^2 / 2\sigma_i^2)},$$

which are symmetrized as $p_{ij} = (p_{j|i} + p_{i|j})/2N$. In the low-dimensional space, similarities are modeled using a Student- t distribution

$$q_{ij} = \frac{(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|\mathbf{y}_k - \mathbf{y}_l\|^2)^{-1}}.$$

The embedding is obtained by minimizing the Kullback–Leibler divergence

$$\mathcal{L}_{\text{t-SNE}} = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}},$$

which encourages samples that are close neighbors in the original space to remain close in the projected space.

In the case of the Multi-Scale Dual-Axis Conformer, the resulting projections reveal tighter intra-class clusters and clearer boundaries between classes, indicating improved local consistency and reduced feature ambiguity. This suggests that the proposed architecture learns embeddings in which samples of the same class occupy more coherent regions of the latent space. Conversely, CTNet exhibits more diffuse and overlapping clusters, implying weaker local discrimination.

In our setting, each point corresponds to a trial-level embedding derived from EEG-based motor imagery data after multi-scale spatiotemporal processing. Since these embeddings are extracted immediately before the classification layer, they can be interpreted as lying on a lower-dimensional latent manifold encoding task-relevant neural patterns. Improved cluster compactness in the t-SNE visualization therefore reflects better local organization of this manifold, consistent with the quantitative performance gains reported in the main paper.